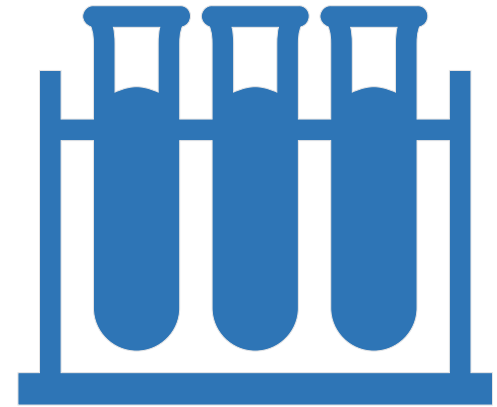
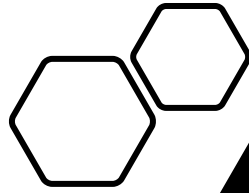


F-tests, ANOVA,
and more on
designs



Mapping in Experimental Design



One-Way Design

- Experiments with only one variable manipulated.

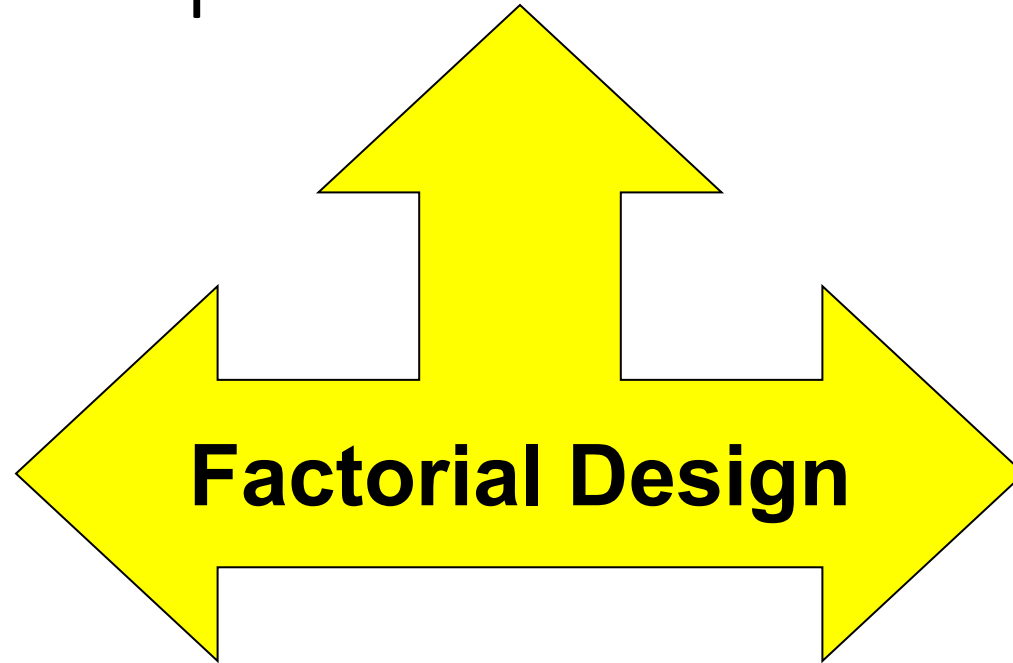
Reading Program Study

Phonics

Whole

Word

A bit more complicated...



Two or more IV manipulated.

Reading & Class Size Study

	Sm	Lg
Phonics		
Whole Word		

Factorial Nomenclature

- Special terms to describe the size and structure of factorial designs.
- A “2 x 2 factorial” (read “2-by-2”) is a design with two independent variables, each with two levels.
- A “3 x 3 factorial” has two independent variables, each with three levels.
- A “2 x 2 x 4 factorial” has three independent variables, two with two levels, and one with four levels.

Reading & Class Size Study

	Sm	Med	L
Phonics			
Whole Word			

Nomenclature: 2 x 3 Factorial Design

How to Assign Subjects?

	Sm	Med	Lg
Phonics	?	?	?
Whole Word	?	?	?



4 ways to assign subjects

1. Randomized Groups (between Ss)
 2. Matched Subjects (between Ss)
 3. Repeated-Measures (within Ss)
 4. Mixed Factorial (within & between Ss)
- 

Reading & Class Size Study

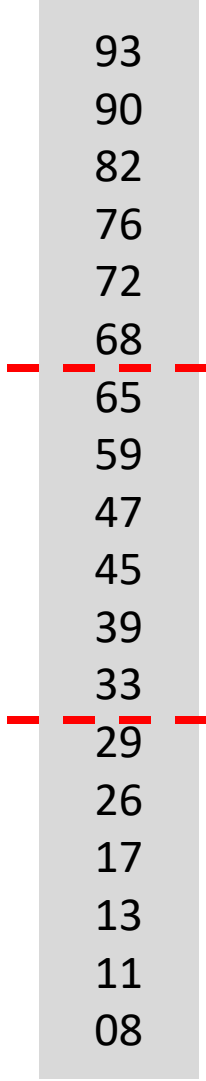
	Sm	Med	Lg
Phonics	Anna, Frank, Beia	Joe, Tim, Marcel	Lauren, Jim, Emily
Whole Word	Alan, Julie, Brooke	Betty, Sarah, John	Ethan, James, Lee

2 x 3 Between Subjects Design

Matched Ss Design

SES Index

- rank order scores from highest to lowest.
- create blocks of subjects with 6 highest scores in first block, 6 next highest scores in next block, etc....
- randomly assign subjects from each block to the groups.



93
90
82
76
72
68
65
59
47
45
39
33
29
26
17
13
11
08

Reading & Class Size Study

	Sm	Med	Lg
Phonics	Lois (93) Frank (65) Beia (29)	Joe (90) Tim (59) Marcel (11)	Lauren (76) Jim (47) Emily (8)
Whole Word	Alan (68) Julie (33) Brooke (13)	Betty (82) Sarah (39) John (26)	Ethan (72) James (45) Lee (17)

2 x 3 Between Subjects Design Matched on SES

Sleep & Alcohol Study – Repeated Measures

	6g	2g	0g
Sleep	Lois Beia Eiden	Lois Beia Eiden	Lois Beia Eiden
Sleep Deprived	Lois Beia Eiden	Lois Beia Eiden	Lois Beia Eiden

Repeated Measures 2 x 3 Factorial Design

Mixed Factorial

	6g	2g	0g
Sleep	Lois Beia Eiden	Betty Ethan Faustino	Paul M Paul H Paul S
Sleep Deprived	Lois Beia Eiden	Betty Ethan Faustino	Paul M Paul H Paul S

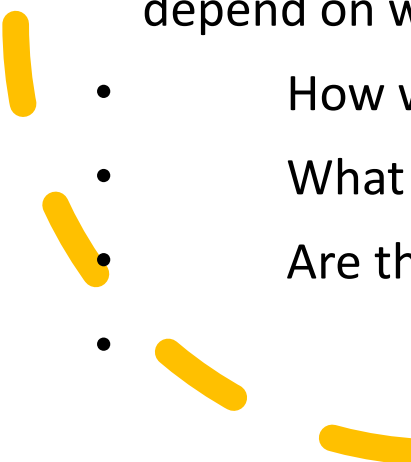
Mixed Factorial Design: 2 x 3 with Alcohol as the Between Ss variable and Sleep Deprivation as the Within Ss variable.

What are the hypotheses?

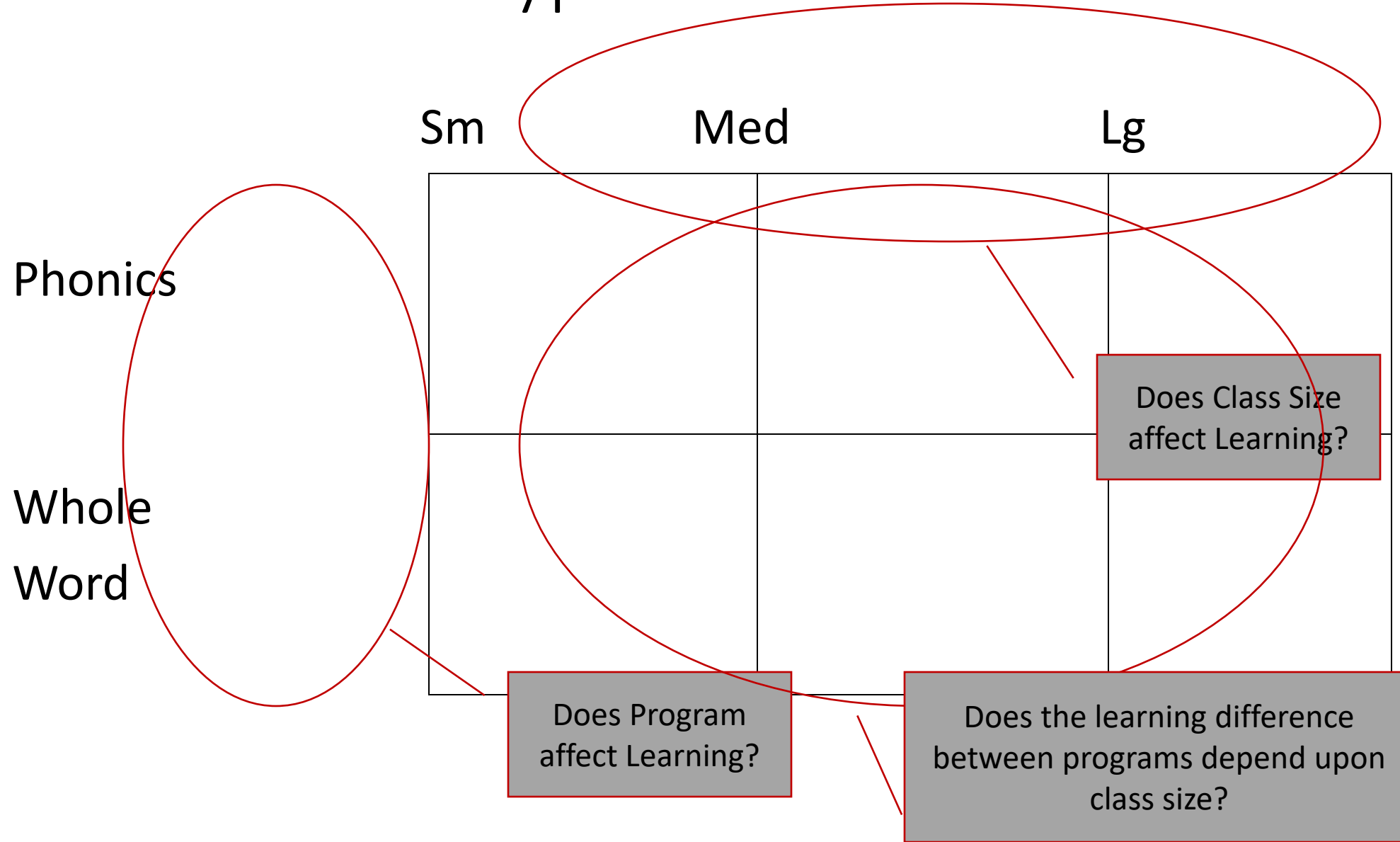
	Sm	Med	Lg
Phonics			
Whole Word			



• Factorial Nomenclature

- How many conditions are in the simplest possible experiment?
 - How many IV are involved in a 3x3 factorial design?
 - A researcher wants to compare the effectiveness of cognitive behavioral therapy and client-centered therapy. And, if the clients are married couples, does the efficacy of their therapy depend on whether spouses attend session together or separate.
 - How would you describe this study?
 - What type of assignment would you use?
 - Are there any variables to match for assignment & why?
 -
- 

What are the hypotheses?





Hypotheses in Factorial Designs

Whenever there are 2 or more IV, there are always at least 3 hypotheses.

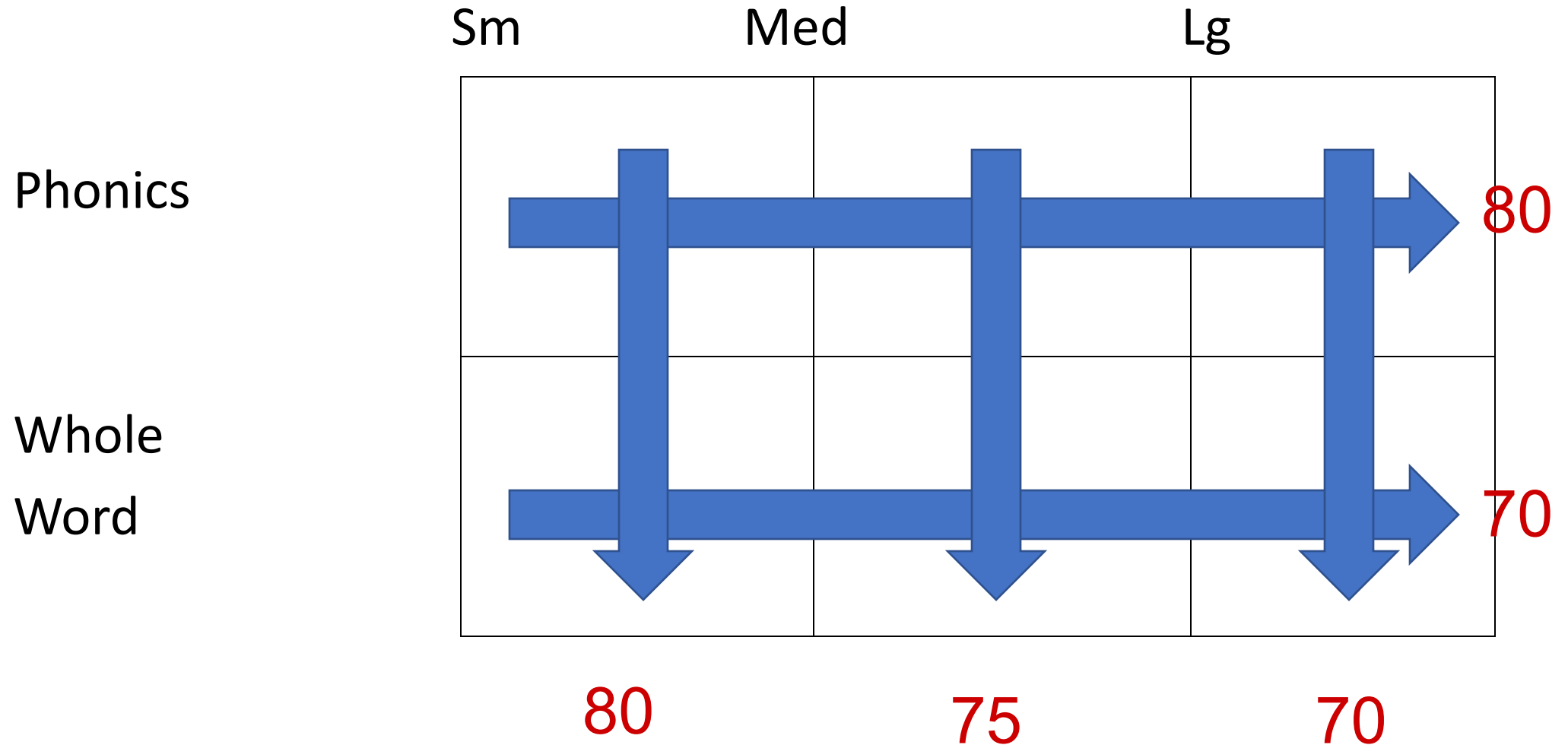
1. Main Effect of IV 1 (Does reading program affect learning?)
2. Main Effect of IV 2 (Does class size affect learning?)
3. Interaction of IV 1 & IV 2 (Does the learning from reading programs differ with class size?)

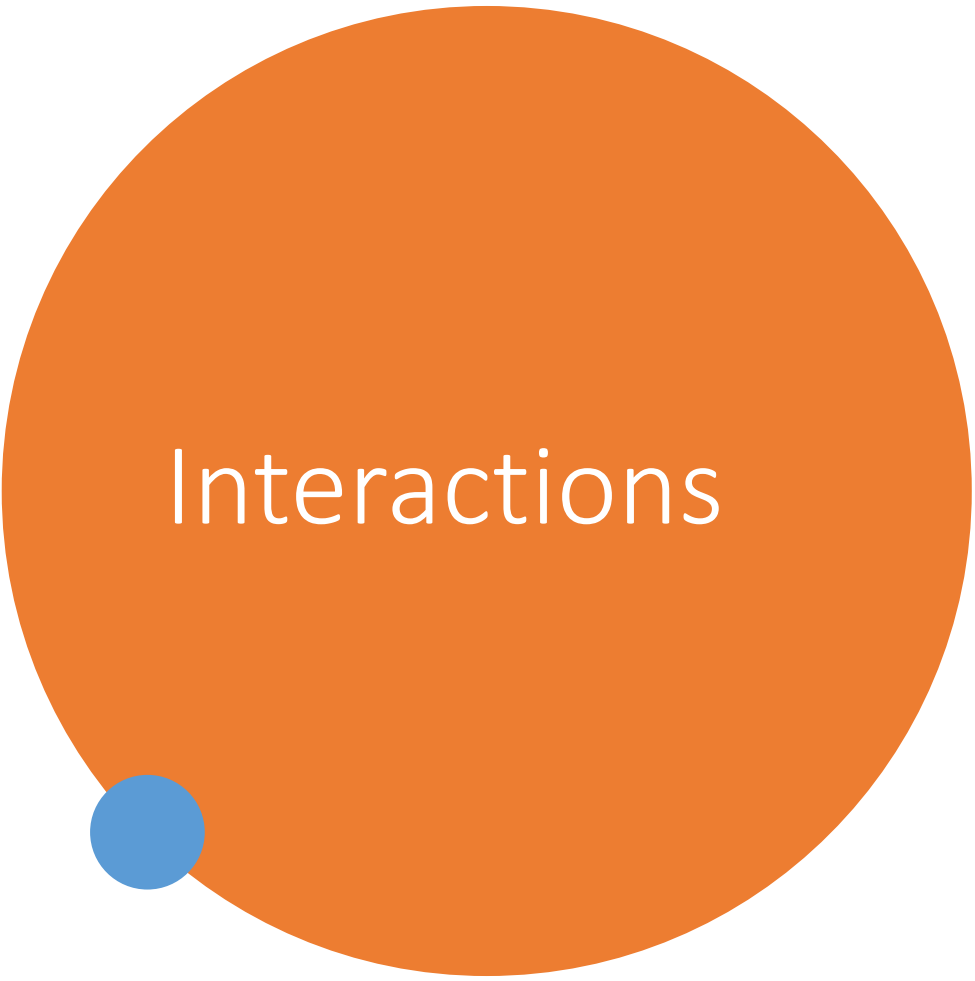
Note: All effects are independent of one another.



Main Effects

Difference
between the levels
of each IV





Interactions



AN interaction between two factors exists if the effect of one factor depends on the level of a 2nd factor.

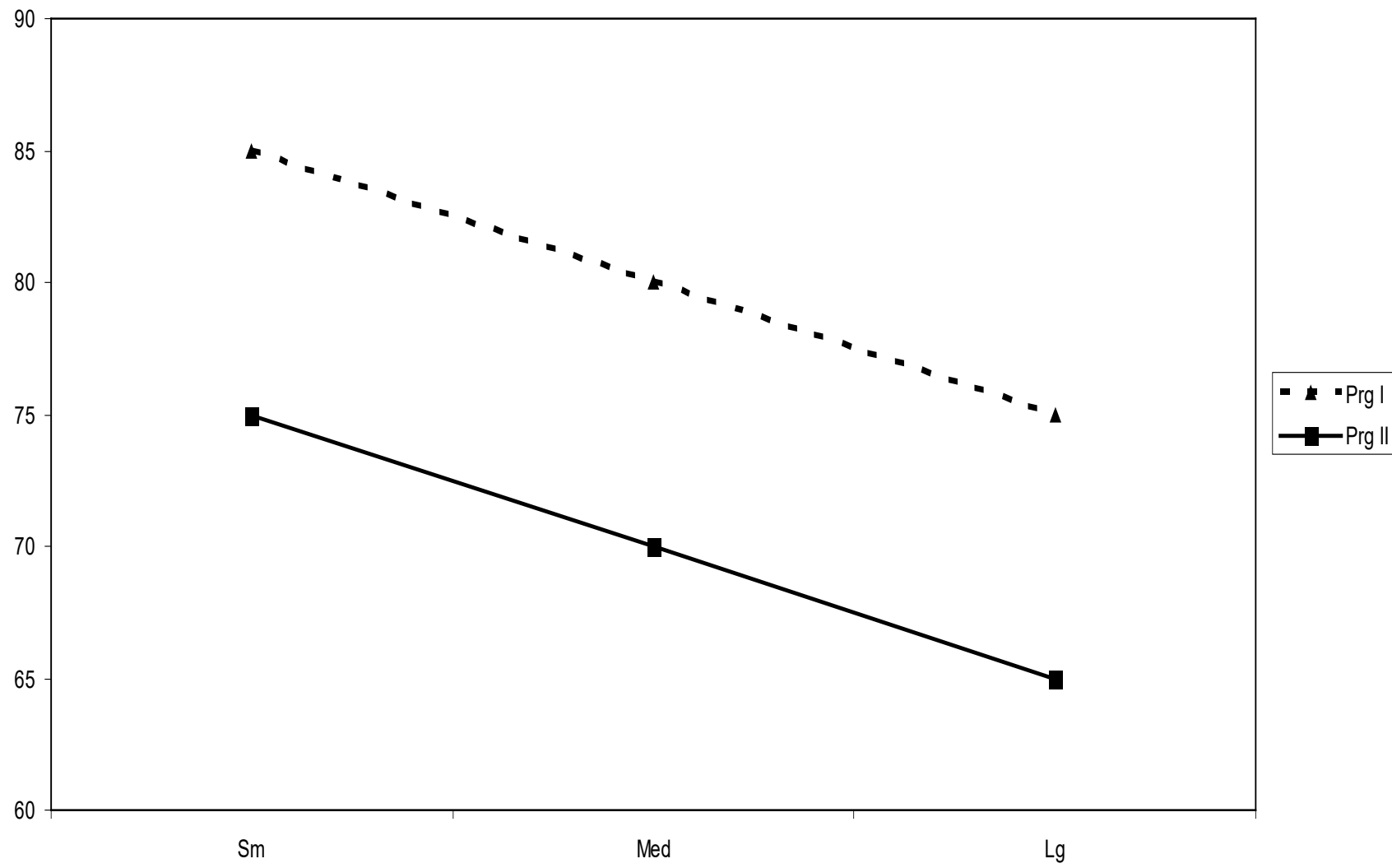
Is there an interaction?

no

	Sm	Med	Lg
Phonics	85	80	75 80
Whole Word	75	70	65 70

10 10 10

~~80~~ ~~75~~ ~~70~~



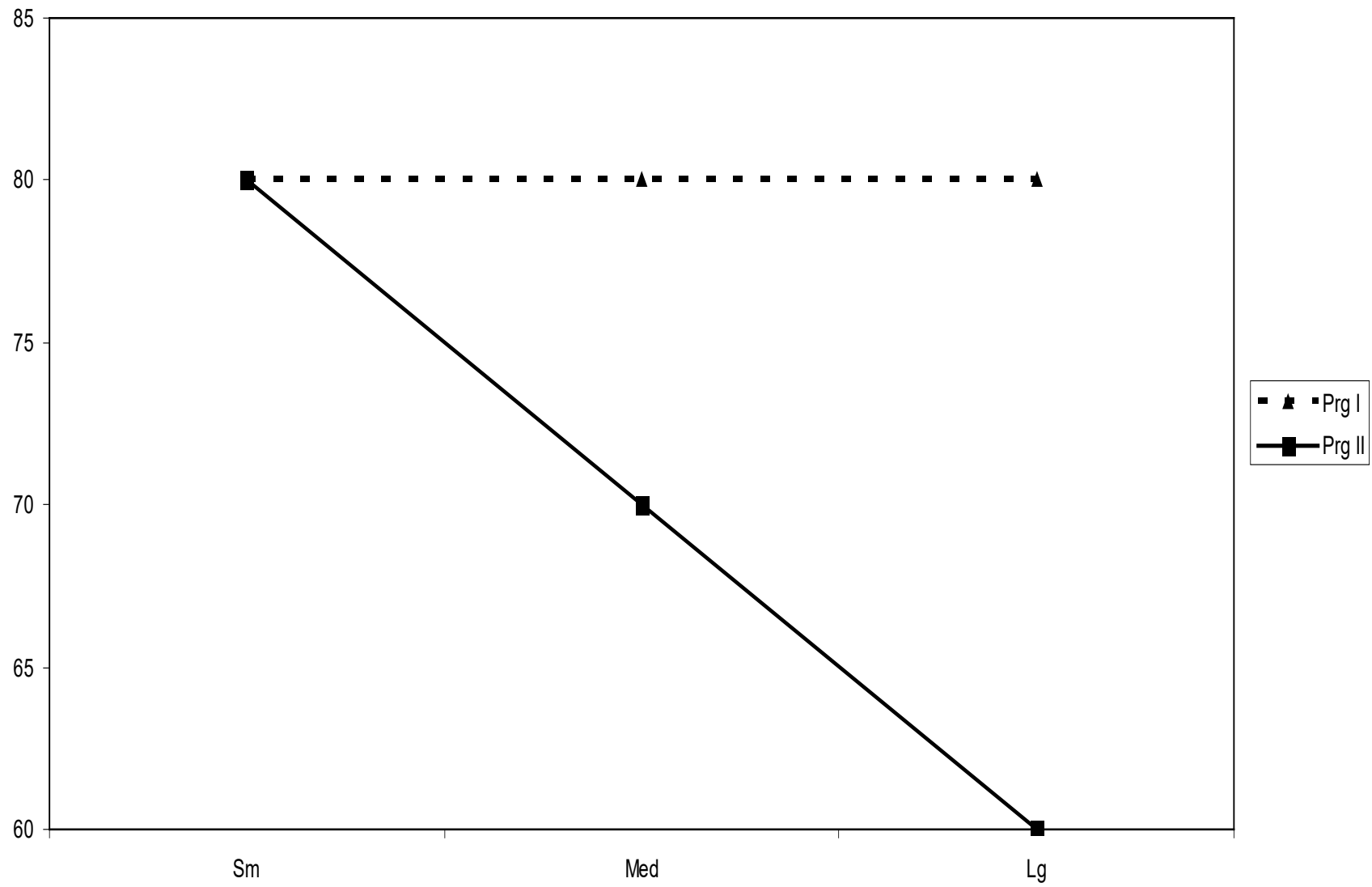
Back to the 1st
grade classroom,
Let's change these
numbers...



Is there an interaction?

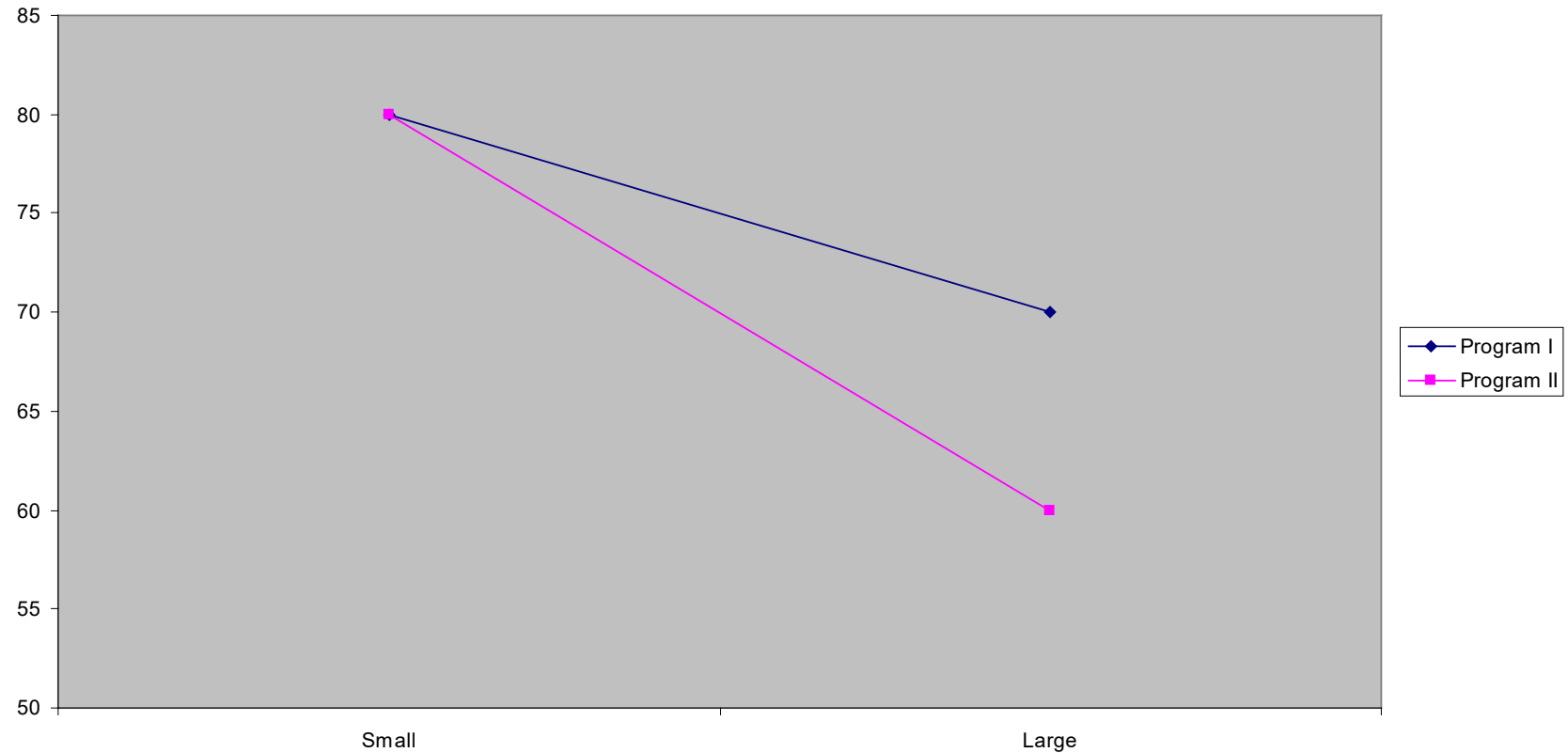
YES

	Sm	Med	Lg
Phonics	80	80	80
	0	10	20
Whole Word	80	70	70
	80	75	70



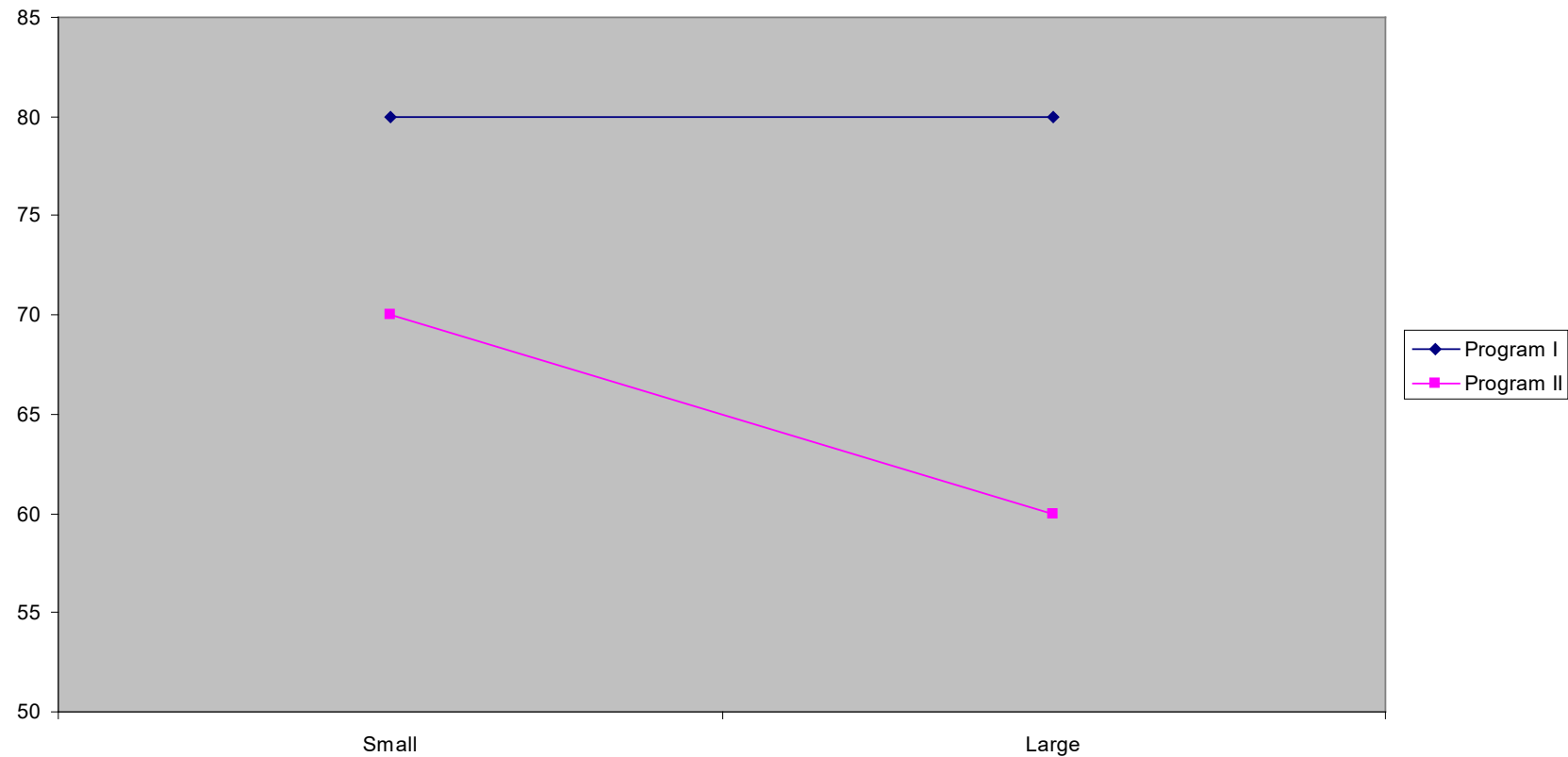
Types of Interactions

1. The Effect of IV at one level is different, but not at the other level.



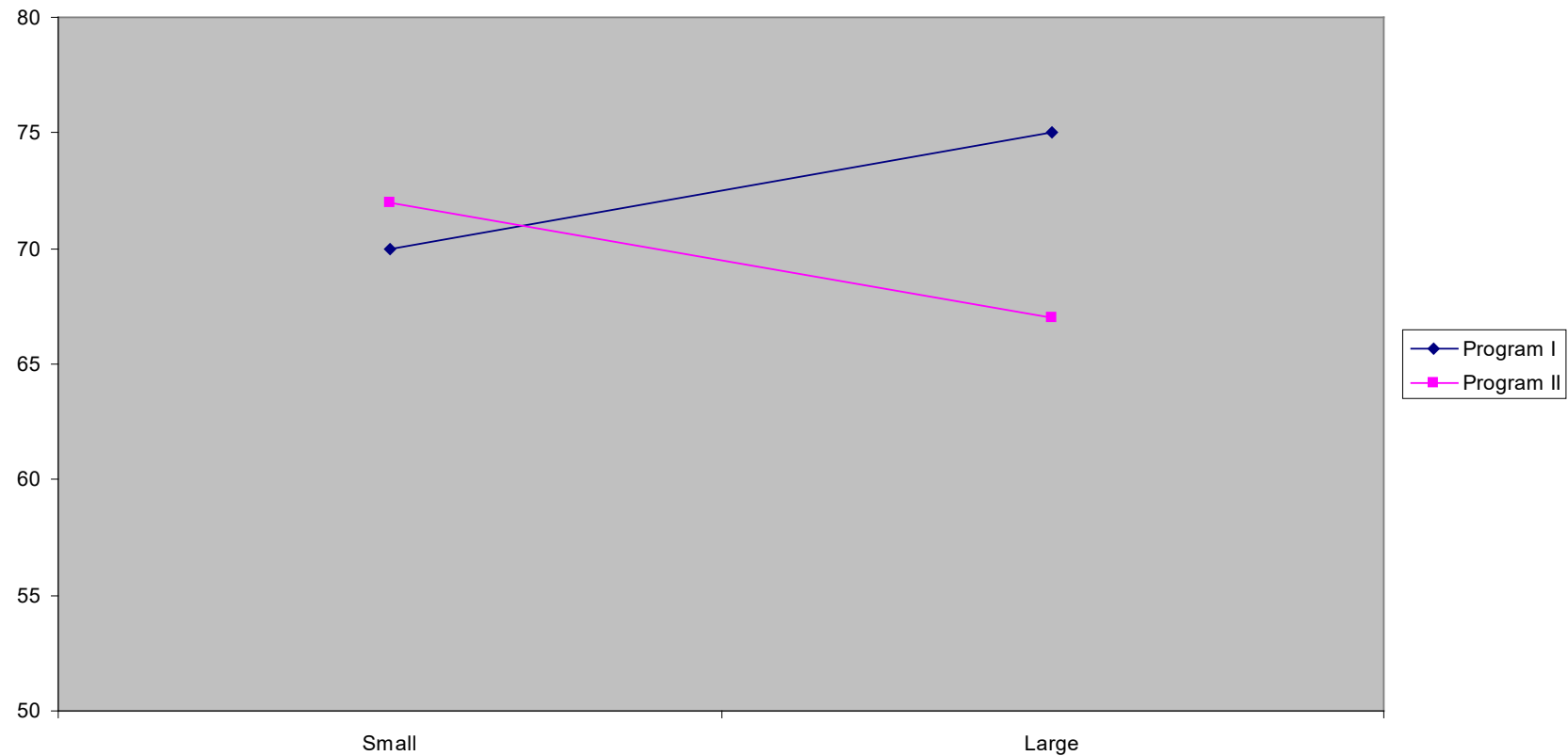
Types of Interactions

2. Effect of IV is same direction for all levels of the other IV, but the effect is stronger at one level than at another.



Types of Interactions

3. The Crossover – the effect of the IV goes in different directions for different levels of the IV.





Wrapping
Up...

Bringing it
together



Basic Workflow for ANOVA

- 4 STEPS
- State the hypothesis and set the criterion ($H_0 : \mu_1 = \mu_2 = \mu_3, \text{ etc })$
- Compute the test statistic $F = MSB / MSW$.
- Compute the p-value. (Need to find the degrees of freedom.)
- Make a decision.



Example

- A college professor wants to determine the best way to present an important lecture topic to their class.
- They decide to do an experiment to evaluate three options.
- They solicit 27 volunteers from their class and randomly assigns 9 to each of three conditions.



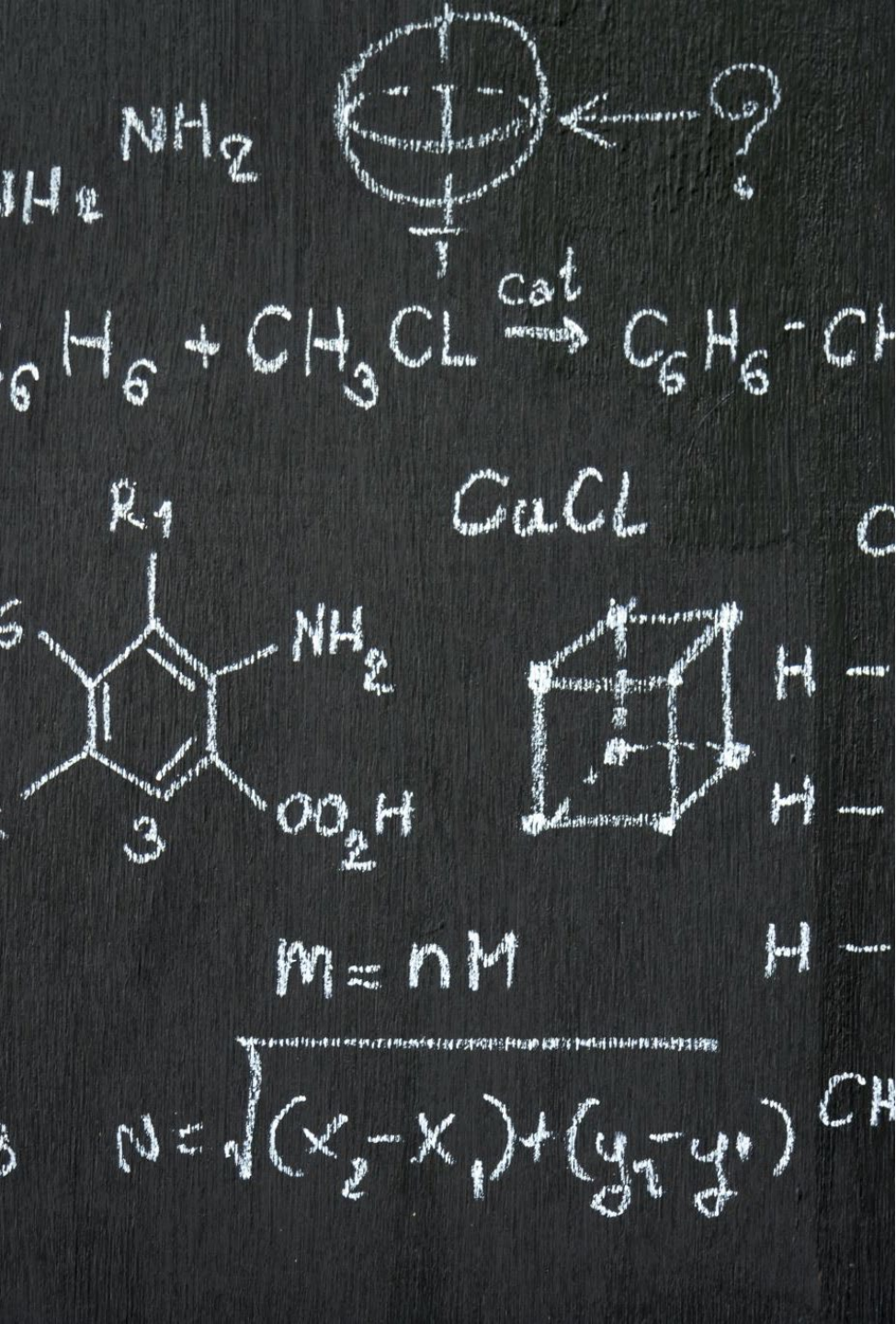
Design

- In condition 1,
 - they lecture to the students.
- In condition 2,
 - they lecture plus assign supplementary reading.
- In condition 3,
 - the students see a film on the topic plus receive the same supplementary reading as the students in condition 2.
- The students are subsequently tested on the material.
- The following scores (percentage correct) were obtained.

Data

- No one does the calculations by hand.
- Always use a computer.

LECTURE CONDITION 1	LECTURE + READING CONDITION 2	FILM + READING CONDITION 3
92	86	81
86	93	80
87	97	72
76	81	82
80	94	83
87	89	89
92	98	76
83	90	88
84	91	83



Step 1) Hypotheses

- for one-way ANOVA, the hypotheses are
- $H_0 : \mu_1 = \mu_2 = \mu_3$
- $H_a : \mu_i \neq \mu_k \text{ for some } i, k$
- Set $\alpha = 0.05$

Step 2) TEST STATISTIC

- Use an on-line calculator...
(like
<https://www.socscistatistics.com/tests/>)
- <https://www.socscistatistics.com/tests/anova/default2.aspx>

Treatment 1	Treatment 2	Treatment 3	Treatment 4	Treatment 5
92	86	81		
86	93	80		
87	97	72		
76	81	82		
80	94	83		
87	89	89		
92	98	76		
83	90	88		
84	91	83		

Step 2) TEST STATISTIC

- Use an on-line calculator...
(like <https://www.socscistatistics.com/tests/>)
- <https://www.socscistatistics.com/tests/anova/default2.aspx>

Summary of Data						
	<i>Treatments</i>					
	1	2	3	4	5	Total
N	9	9	9			27
ΣX	767	819	734			2320
Mean	85.2222	91	81.5556			85.926
ΣX^2	65583	74757	60088			200428
Std.Dev.	5.2148	5.3385	5.3177			6.4446

Results

- We read out the results of the analysis in the ANOVA summary table
- Lots of Info in here

Result Details				
<i>Source</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	
Between-treatments	408.0741	2	204.037	$F = 7.28945$
Within-treatments	671.7778	24	27.9907	
Total	1079.8519	26		

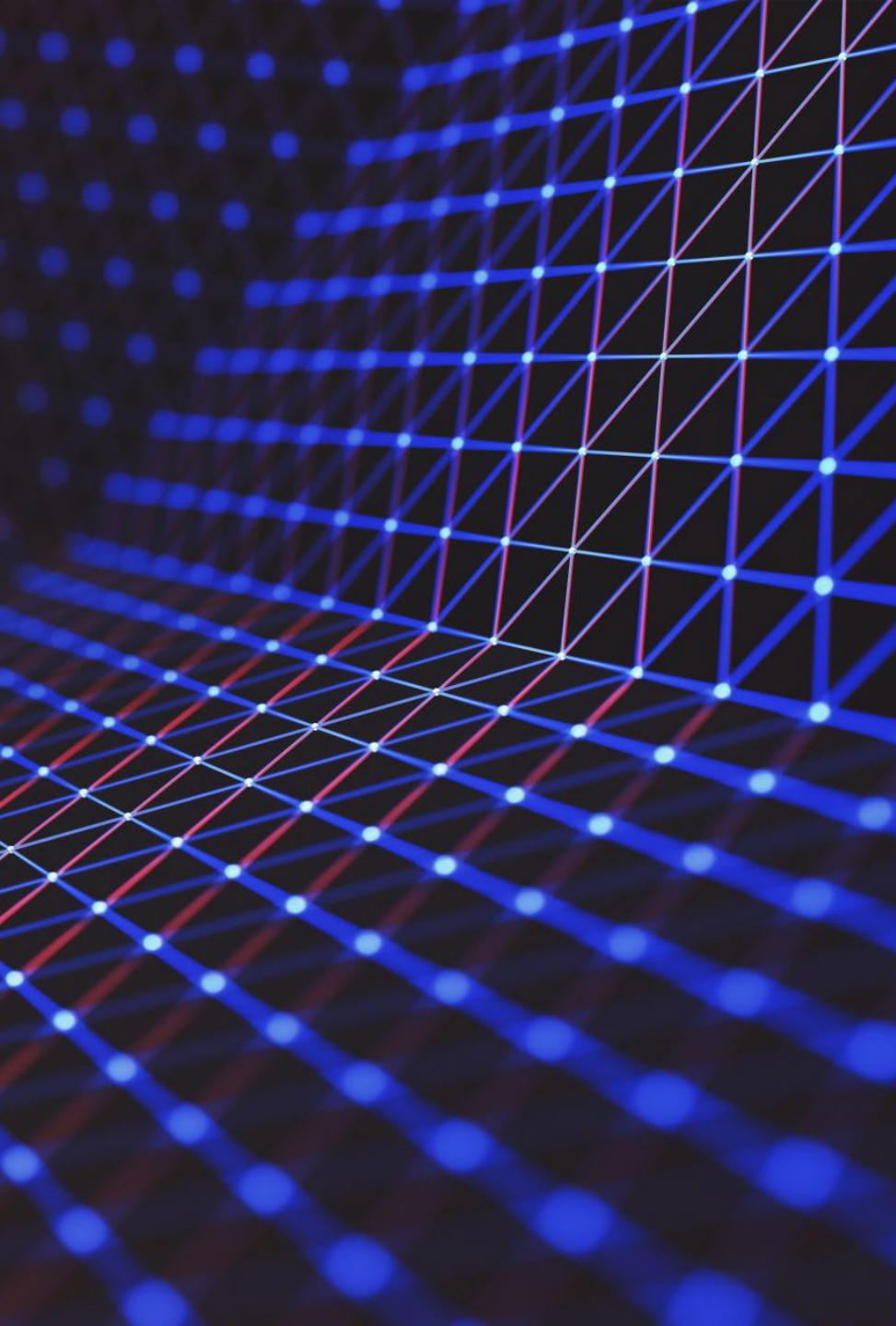
The F -ratio value is 7.28945. The p -value is .00336. The result is significant at $p < .05$.

We can double check things

- $F = MSB/MSW =$
- $204.0370/27.9907 = 7.2894$
- $MSB = SSB / (K-1) = 408.0741 / (3-1) = 204.0370$
- $MSW = SSW / (N-K) = 671.7778 / (27-3) = 27.9907$

Result Details				
Source	SS	df	MS	
Between-treatments	408.0741	2	204.037	$F = 7.28945$
Within-treatments	671.7778	24	27.9907	
Total	1079.8519	26		

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Step 3) p VALUE

between degrees of freedom (numerator)

- $df = K - 1 = 3 - 1 = 2$

within degrees of freedom (denominator)

- $df = N - K = 27 - 3 = 24$

Total degrees of freedom

- $df = N - 1 = 27 - 1 = 26$

Check the p-value using an F distribution calculator

- <https://www.socscistatistics.com/pvalues/fdistribution.aspx>

P-Value from F-Ratio Calculator (ANOVA)

This should be self-explanatory, but just in case it's not: your *F*-ratio value goes in the *F*-ratio value box, you stick your degrees of freedom for the numerator (between-treatments) in the *DF*- numerator box, your degrees of freedom for the denominator (within-treatments) in the *DF*- denominator box, select your significance level, then press the "Calculate" button.

If you need to derive an *F*-ratio value from raw data, [you can find an ANOVA calculator here](#).

F-ratio value:

DF- numerator:

DF- denominator:

Significance Level:

- ☐ .01
- ☒ .05
- ☐ .10

4) Decide

- Because $p = 0.00336 < .05$
- we reject H_0 .
- The methods of presentation are not equally effective.
- Note, does not tell us which pair of means are different! Look at means
- Look at means!

Step 2) TEST STATISTIC

- Use an on-line calculator...
(like <https://www.socscistatistics.com/tests/>)
- <https://www.socscistatistics.com/tests/anova/default2.aspx>

Summary of Data						
	<i>Treatments</i>					
	1	2	3	4	5	Total
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ΣX^2	65583	74757	60088			200428
Std.Dev.	5.2148	5.3385	5.3177			6.4446



Big Idea

- The great thing about ANOVA is that
 - these basic steps stay the same even if you have many more means to be compared.
- I could just happen to have data from 8 different classes
 - that all completed an experiment
 - where subjects responded as quickly as possible whether a set of letters formed a word or not
- The summary is the same format as above 😊

Basic Workflow for ANOVA

4 STEPS

State the hypothesis and set the criterion
($H_0 : \mu_1 = \mu_2 = \mu_3, \text{ etc })$

Compute the test statistic $F = \text{MSB} / \text{MSW}$
.

Compute the p-value. (Need to find the
degrees of freedom.)

Make a decision.



Wrapping up!

